

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (CCE) (Credit System) Degree Examinations
 January 2023

20CC512 – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I

	Marks	BT*	CO*	PO*
1. a) What is rational agent? Explain "Rationality" with an example vacuum cleaner.	5	L*2	1	1
b) Explain ANY FIVE types of task environments that might arise in AI. Mention suitable example for each of them.	10	L 2	1	1
c) With schematic diagram, explain goal based agent program.	5	L 2	1	1
2. a) Explain ANY FIVE branches of Artificial Intelligence.	5	L 2	1	1
b) Outline state of the art of Artificial Intelligence.	10	L 2	1	1
c) Explain "Toy Problem" by considering vacuum cleaner as an example. Clearly mention the five components of problem formulation.	5	L 2	1	1
3. a) Define TASK environment and explain the TASK for automated taxi driver.	5	L 2	1	1
b) Illustrate agents and environment by taking "Vacuum Cleaner" as an example.	10	L 2	1	1
c) With schematic diagram, explain model based agent.	5	L 2	1	1

Unit – II

4. a) Define data preprocessing. List different types of data preprocessing techniques. With a flowchart, explain feature subset selection process. 8 L 2 2 2

b) What is Naive Bayes classifier? With an example, explain how class label can be predicted using Naive Bayes classification technique. 6 L 2 4 2

c) A database has 5 transactions. Let min_sup = 3. Find all frequent itemsets by applying Apriori Algorithm.

TID	Items bought
T100	E, A, R, I, N, G
T200	D, A, R, I, N, G
T300	E, P, I, N
T400	E, U, C, I, G
T500	C, A, A, I, H, N

6. a) Explain Maximum margin hyperplanes concepts applied in SVM classifier. 5 L 2 4 2

- b) Trace the results of applying the Apriori algorithm on the grocery store example with support threshold $s=33.34\%$ and confidence threshold $c=60\%$. Show the candidate and frequent itemsets for each database scan. Enumerate all the final frequent itemsets. Also indicate the association rules that are generated and highlight the strong ones, sort them by confidence.

Transaction ID	Items
T1	HotDogs, Buns, Ketchup
T2	HotDogs, Buns
T3	HotDogs, Coke, Chips
T4	Chips, Coke
T5	Chips, Ketchup
T6	HotDogs, Coke, Chips

10 L 3 3 2

- c) Explain the characteristics of Nearest-Neighbor Classifiers.

5 L 2 4 2

6. a) Explain three types of machine learning types. Mention two algorithms for each of the types.

6 L 2 2 2

- b) A database has the following transactions. Assume support threshold = 50%. Generate the frequent itemsets by applying Apriori algorithm.

TID	Items bought
T1	I1, I2, I3
T2	I2, I3, I4
T3	I4, I5
T4	I1, I2, I4
T5	I1, I2, I3, I5
T6	I1, I2, I3, I4

8 L 3 3 2

- c) Explain decision tree induction for classification.

6 L 2 4 2

Unit – III

7. a) Explain K-means clustering algorithm and its working with an example.

10 L 2 5 2

- b) Explain DBSCAN clustering algorithm.

10 L 2 5 2

8. a) What is BIRCH algorithm? Describe all the phases of BIRCH algorithm.

10 L 2 5 2

- b) Explain different types of clusters.

10 L 2 5 2